

We don't try to do details here: They are "too non-linear for this book." However, a recent explicit variational principle for circle packing found by Bobenko and Springborn [109] yields a very elegant proof for Theorem 4.13. This is worked out in detail in [580]; for the uniqueness part see also Springborn [510].

Still a different polytope realization algorithm, for simplicial 3-polytopes, was given by Das & Goodrich [178]: it essentially works by doing "many inverse Steinitz operations on independent vertices in one step," and thus yields a linear-time realization algorithm that produces realizations with a singly-exponential bound on the size of the integer coordinates (as does the Onn–Sturmfels algorithm [428] [459, Sect. 13.2] for general 3-polytopes).

Elementary semialgebraic sets are basic objects. Exercise 4.22 indicates that they can have very complicated structure. General semialgebraic sets (for which also nonstrict inequalities are admitted in the defining system) can be written as finite unions of elementary semialgebraic sets.

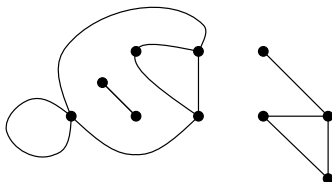
In this chapter we only met realization spaces that are quite trivial (contractible); this will change as soon as we have developed the theory to analyze the realization spaces of some high-dimensional polytopes, in Lecture 6. The study of semialgebraic sets is called real algebraic geometry — a very active and fascinating field of research. We refer to Bochnak, Coste & Roy [110] and Becker [65] for more information.

Problems and Exercises

- 4.0 Show that the complete graph K_5 , and the complete bipartite graphs $K_{3,n}$, are ΔY -reducible. Considering the graphs that are ΔY -equivalent to K_6 , show that K_n is not ΔY -reducible, for $n \geq 6$, and that $K_{m,n}$ is not ΔY -reducible for $m, n \geq 4$.
- 4.1 If G is a 3-connected graph with $n \geq 4$ edges, show that it contains a subdivision of K_4 .
- 4.2 Show that the definition of k -connectivity given in the notes to this lecture specialize to our definitions in Section 4.1 of 2-connectivity (for graphs with more than 1 edge) and of 3-connectivity (for graphs with more than 3 edges).
- 4.3 What is the problem with the construction of P from P' , if $G \longrightarrow G'$ is a simple Y -to- Δ transformation? For this, try to prove the Y -to- Δ part of Lemma 4.3.
If you are comfortable with projective transformations (Section 2.7), explain how to do this without using polarity (as we did).
- 4.4 Check how to do the reduction theorem for grid graphs entirely within the framework of 3-connected graphs. Where are the problems? How

do you make grid graphs 3-connected? How many “basic operations” do you need?

- 4.5 Characterize the graphs of centrally symmetric 3-polytopes. (Grünbaum [252, Thm. 13.2.5])
- 4.6 Let G be any finite graph drawn into the plane without crossings, and let v be its number of vertices, e its number of edges, c its number of connected components, and f the number of connected regions determined by it. For example, the graph



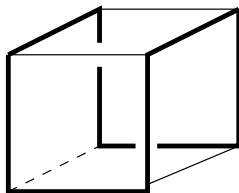
has $v = 10$, $e = 11$, $f = 5$, and $c = 3$.

Show that in general $v - e + f = 1 + c$. Deduce *Euler's formula*

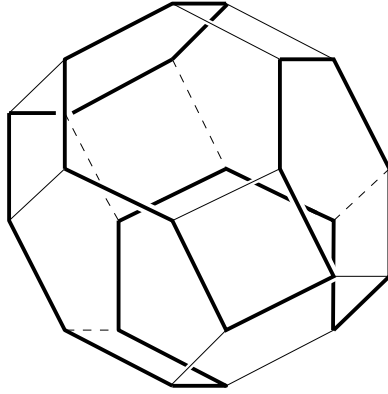
$$v - e + f = 2$$

for the number of vertices v , the number of edges e , and the number of facets f of a 3-polytope.

- 4.7 Deduce from Steinitz' theorem (from Corollary 4.9) that *every* planar graph has a straight line drawing in the plane without intersections. (In graph theory literature, this appears as a result of Wagner [554], rediscovered by Fáry [205]. It is also quite easy to prove this directly by induction on the number of vertices; see, for example, Hartsfield & Ringel [273, p. 167].)
- 4.8 Conclude from Exercise 4.6 that for every 3-polytope P , either the polytope P itself or its polar P^Δ has a facet that is a simplex. (This is not true for 4-polytopes — for these read about the regular 24-cell, whose facets are octahedra and whose vertex figures are cubes; see, for example, Coxeter [164, Sect. 8.2].)
- 4.9 The graph of the 3-cube, $G(C_3)$, has cycles that go through all the vertices (“Hamiltonian cycles”).

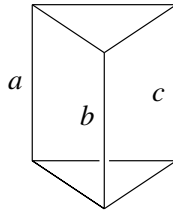


Take such a cycle, and then construct a realization of a combinatorial cube in 3-space such that the usual projection $\pi : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ carries the cube to an 8-gon, and the cycle to the boundary cycle of the 8-gon. Then do the same for the 3-dimensional permutahedron Π_3 ,



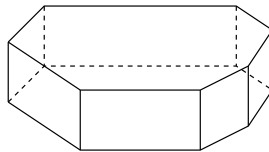
and for the (regular) dodecahedron.

- 4.10 Show that we cannot prescribe the shape of two facets of a 3-polytope (even if they have congruent intersection). In fact, in a triangular prism



if one square face is prescribed such that the edges a and b are parallel, then the other square faces have to have b and c respectively a and c parallel.

- 4.11 One cannot prescribe two disjoint facets of a 3-polytope either. For this, analyze the prism over an n -gon, and show the following. If we prescribe the bottom n -gon, then the coordinatizations of the completed prism are prescribed by $n + 3$ linear parameters (“degrees of freedom”), while we have $2n - 3$ choices for the shape of the top facet. Thus, if we prescribe two generic n -gons, then they cannot be built into a prism, if $n \geq 7$.



- 4.12 Show that we cannot prescribe the shape of the shadow boundary of a 3-polytope.

(Jürgen Richter-Gebert, who noted that from a triangular prism, you can get a hexagon as a projection, but the shape of the hexagon cannot be prescribed: see page 141! Earlier, Barnette [46] gave a proof for the case where the 3-polytope is a tetrahedron with a stellar subdivision on every facet, and Q is a regular 8-gon. Also, you can use the prisms of the previous exercise, and again count degrees of freedom.)

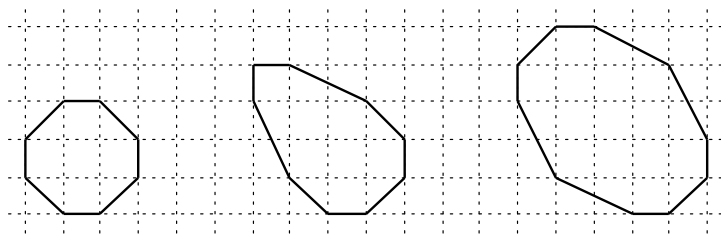
- 4.13 Show that not all 3-polytopes can be represented in such a way that all facets touch the unit sphere, or that all vertices are on the unit sphere. Which 3-polytopes can be represented this way?

(Remark: This is a classical problem that goes back to 1832; see Steiner [523], Steinitz [525], and Schulte [486]. The characterization problem was solved very recently by Hodgson, Rivin & Smith [277].)

- 4.14 Show that if an n -gon is represented in $\mathbb{R}^2$ in such a way that all of its edges touch S^1 in their midpoints, then the n -gon is regular.

If the edges just touch S^1 , but not necessarily with their midpoints, then the n -gon need not be regular, even if we require that the sum of the vectors of the contact points is $\mathbf{0}$.

- 4.15 Let $f_2(n) \in \mathbb{N}$ be the smallest number such that the convex n -gon $P_2(n)$ can be represented on an $f_2(n) \times f_2(n)$ grid, that is, with all vertices in $\{0, 1, \dots, f_2(n)\}^2$. For example, we have $f_2(3) = f_2(4) = 1$, $f_2(5) = f_2(6) = 2$, $f_2(7) = f_2(8) = 3$, $f_2(9) = 4$, but $f_2(10) = 5$ (see the figure).



- (i) Show that $f_2(n) \leq cn^{3/2}$ for some $c > 0$.
(In fact, Thiele [539, Satz 4.1.10] proved that the bound

$$f_2(n) = 2\pi \left(\frac{n}{12}\right)^{3/2} + O(n \log n)$$

is sharp.)

- (ii) Using part (i), show that the cyclic 3-polytope $C_3(n)$ with n vertices can be represented on a $k \times k \times 3$ grid, where $k = f_2(n-1)$.

- (iii) Using part (i), show that the polar cyclic 3-polytope $C_3(m)^\Delta$ with n vertices (where $n = 2m - 4$) can be represented on a $k' \times k' \times k'$ grid, where $k' = f_2(\frac{n}{2} + 1)$.

4.16* For $n \geq 4$ let $f_3(n)$ be the smallest positive integer such that every 3-dimensional polytope with n vertices can be represented with integral vertices in $\{0, 1, \dots, f_3(n)\}^3$. Let $f_3^s(n)$ be the same function for simplicial 3-polytopes.

Determine the asymptotic behavior of the functions $f_3(n)$ and $f_3^s(n)$.

(Work by Goodman, Pollack & Sturmfels [238, Sect. 5] shows that for d -dimensional simplicial polytopes with $d + 4$ vertices one needs vertex coordinates that grow doubly exponentially in d ,

$$f_d^s(d + 4) \geq 2^{2^{cd}}$$

for some constant $c > 0$. However, there is no such result that applies to *simplicial* polytopes for any constant dimension d .

The case $d = 3$ might be special: it is quite possible that there is a quadratic upper bound on $f_3(n)$; up to now only exponential upper bounds $f_3^s(n) \leq 28.45^n$ and $f_3(n) \leq 533^{n^2}$ are known, due to Onn & Sturmfels [428], Richter-Gebert [459, p. 143], and finally Ribó Mor & Rote [454, Chap. 6].

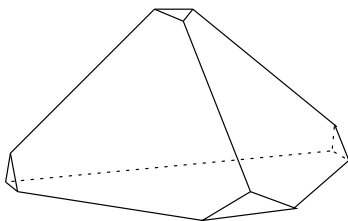
4.17* For $n \geq 4$, let $g(n) \in \mathbb{N}$ be the smallest number such that every 3-connected planar graph with n vertices can be represented in the plane in such a way that the vertices are on the $g(n) \times g(n)$ grid, the edges are straight, and the bounded regions determined by the graph embedding, as well as the union of all these regions, are strictly convex (i.e., all interior angles are smaller than π). How large is $g(n)$?

(If one just requires straight-line embeddings, without the convexity assumptions, then one can embed the graph on a $(2n-4) \times (n-2)$ grid; see de Fraysseix, Pach & Pollack [209], and even on an $(n-1) \times (n-1)$ grid, by Schnyder [479]. Chrobak & Kant [157] [311, Section 10.2.2] have a convex (but *not* strictly convex) drawing algorithm that runs in linear time and only needs a grid on $(n-1) \times (n-1)$ vertices! Note that the superlinear lower bounds of Problem 4.15 apply here. With the strict convexity assumption, the currently best upper bound is $O(n^3) \times O(n^3)$, due to Chrobak, Goodrich & Tomassia [156].)

4.18* Does every 3-polytope have a realization with rational edge-lengths?

4.19* Can every (simple) convex 3-polytope P be represented in such a way that a combinatorially polar polytope $Q \cong P^\Delta$ can be constructed as the convex hull of vertices that are chosen on the corresponding facets of P ?

Can you do this for the 3-polytope obtained by cutting off the vertices of a tetrahedron?



That is, can you represent this polytope in such a way, with points on its facets, that adjacent facets of P correspond to adjacent vertices on the convex hull P of the points?

(For the special polytope P above Jürgen Richter-Gebert saw in 1993 that this can be done. In 2004, Andreas Paffenholz has constructed a POLYMAKE model for it.

The general problem is suggested by the misleading description of polar polytopes in the book by Bartels [55, p. 74]. Grünbaum & Shephard posed the question as Problem 3 in [259]. Grünbaum sent me the following e-mail message:

As far as I know, the problem is still open. I am inclined to believe that the answer is negative, and that once some counterexamples are found, we will all be saying how obvious that is.

I agree.)

4.20 Which subsets of $\mathbb{R}$ are elementary semialgebraic?

4.21 Show that for every convex d -polytope with n -vertices, the realization space is an elementary semialgebraic set in $\mathbb{R}^{d \times n}$.

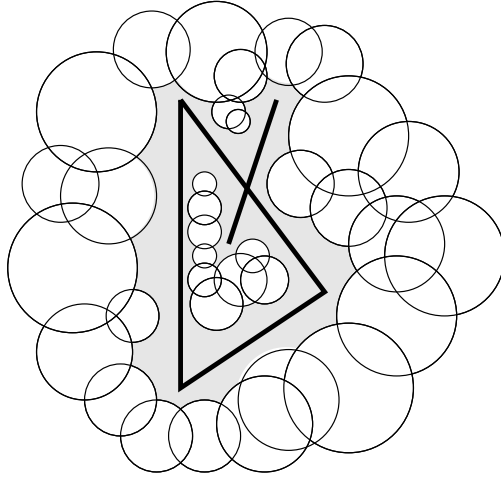
(For this, embed the polytope into $\mathbb{R}^{d+1}$, consider the maximal determinants of a realization matrix, and note that the condition that $d+1$ points have to be on a common facet says that a certain subdeterminant is 0. Similarly, two points being on the same side of the facet spanned by some basis means that the product of two determinants is positive.)

4.22 Let $S \subseteq \mathbb{R}^d$ be a polyhedral set: a finite union of convex polytopes.

Show that there is an open semialgebraic set $M \subseteq \mathbb{R}^d$ that is a neighborhood for S , such that S is a deformation retract of M .

(Hint: First show that the relative interior, and the exterior, of any ball is an open elementary semialgebraic set. Also the intersection of two elementary semialgebraic sets is elementary semialgebraic. Thus

M can be constructed, for example, by deleting a finite number of small closed balls from a larger open ball.



In the drawing, S is a union of a line segment and the boundary of a triangle; the set M appears shaded.)

4.23 The following problems discuss different ways in which a planar graph can be represented by objects in the plane.

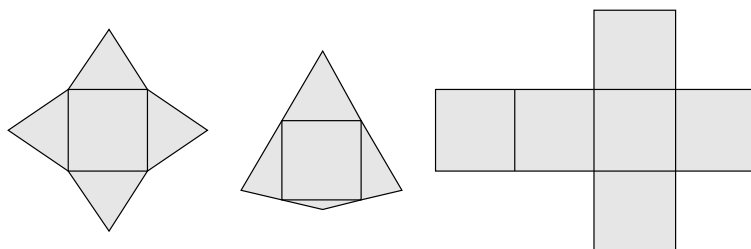
- (i) Prove that the vertices of a planar bipartite graph can be represented by horizontal and vertical closed line segments in the plane, such that the segments intersect if and only if the corresponding vertices are adjacent.
(Hartman, Newman & Ziv [272])
- (ii) In fact, every planar bipartite graph can be represented by disjoint horizontal and vertical open line segments in the plane, which touch if and only if the corresponding vertices are adjacent.
(de Fraysseix, Ossona de Mendez & Pach [429, Sect. 6.3] [208])
- (iii)* Can every planar graph be represented by a family of line segments in the plane such that every vertex corresponds to a segment and adjacent vertices correspond to intersecting edges?

4.24 Describe a “fast” algorithm to test whether two 3-polytopes are combinatorially equivalent.

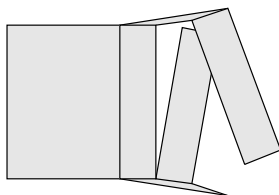
For this, assume that the combinatorial structure is given by the vertex-facet incidence matrices of the polytopes; from these, construct the graphs. Then use that graph isomorphism can be tested “in linear time” for planar graphs, according to Hopcroft & Wong [283].

(As a theoretical concept of *fast algorithms* one has the theory of polynomial algorithms and NP-completeness; see Garey & Johnson [223]. The question of whether a polynomial algorithm exists for isomorphism of general graphs is one of the prominent open problems in this theory [223, pp. 155–158].)

- 4.25* A simple planar representation of a 3-polytope is obtained if we “cut it open” along a spanning tree in its graph, lay out the resulting structure in the plane. Thus we obtain *nets* of 3-polytopes. The following sketches show nets for a symmetric square pyramid, for a not-so-symmetric square pyramid, and for a symmetric cube.



Not every way to open a 3-polytope leads to a valid net without overlaps. For example, the following drawing shows a “bad” way to unfold a 3-polytope that is combinatorially equivalent to the 3-cube.



Can every 3-polytope be represented by a planar net without overlaps?

(This problem appears in Shephard [496], see also [168, Problem B21]. Nets of 3-polytopes were studied extensively in Alexandrow [11]. Combinatorially different 3-polytopes may have the same net — thus we cannot really “represent” 3-polytopes by nets; an example for this was given by Shephard [496]. Another surprising fact in this connection can be found in Namiki, Matsui & Fukuda [420]: even tetrahedra may have overlapping nets!)

- 4.26* Characterize the graphs of 4-dimensional polytopes.

5

Schlegel Diagrams for 4-Polytopes

Now that we understand the combinatorics of 3-polytopes (do we?), the next step is to investigate 4-polytopes. Those are harder to understand, since we (i.e., most of us) lack a genuine geometric intuition for the geometry of 4-dimensional Euclidean space. Nevertheless, there are various tools available. The most prominent one is the “Schlegel diagram” of a polytope, a polytopal complex that represents most of the geometry of a 4-polytope. We will discuss polytopal complexes in some detail (they will be needed for other purposes as well), and then we get to discuss Schlegel diagrams, and some of the traps involved.

5.1 Polyhedral Complexes

Definition 5.1. A *polyhedral complex* $\mathcal{C}$ is a finite collection of polyhedra in $\mathbb{R}^d$ such that

- (i) the empty polyhedron is in $\mathcal{C}$,
- (ii) if $P \in \mathcal{C}$, then all the faces of P are also in $\mathcal{C}$,
- (iii) the intersection $P \cap Q$ of two polyhedra $P, Q \in \mathcal{C}$ is a face both of P and of Q .

The *dimension* $\dim(\mathcal{C})$ is the largest dimension of a polyhedron in $\mathcal{C}$. The *underlying set* of $\mathcal{C}$ is the point set $|\mathcal{C}| := \bigcup_{P \in \mathcal{C}} P$.

$\mathcal{C}$ is a *polytopal complex* if all the polyhedra in $\mathcal{C}$ are bounded (polytopes).